Intelligent Platform for Crowdsourcing

Fall 2022

Introduction

- Summary/Goal of Project
  - Trust between online users (CredRank Algo)
  - Research on FB, Uber, LinkedIn
- Good/High-value Comments
  - Identifying good/bad comments
  - Reasoning behind choosing regression over naive bayes
  - Rewarding Users
  - Encouraging good comments
- Detecting a bot/Flagged Content
  - Types of “bad” comments (bot-generated, short in length, repetitive)
  - Discouraging bad comments
- Visuals
  - Layout of Forum
- Future Plans

Introduction

→ Earlier Projects:

→ Polling Application

→ Automatic Hashtag Generator

→ TLDR Generator

→ Supporting Challenging Conversations Online

→ Difficulties in trusting other users

→ Lack of motivation to participate in thorough discourse online

Sub-Teams

Trustworthiness

- Establishing trust between users online

Identify

- Identifying good vs. bad comments

Encourage

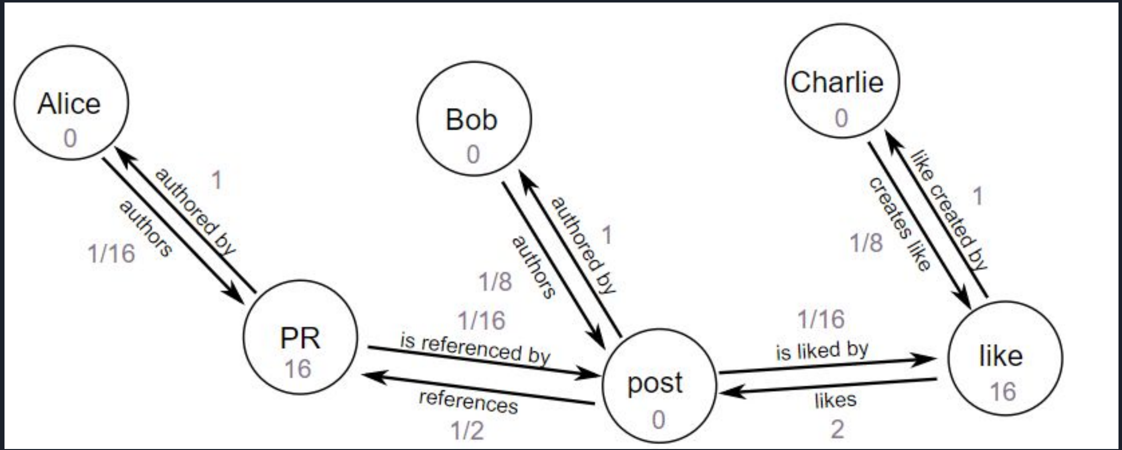
- Extracting high value discourse in forum participants

How Major Companies Utilize Trust

- Uber/Lyft/AirBnb
  - Rating system between service provider and customer.
  - Some form of Insurance
  - Location Tracking
  - Background Checks
- Social Media (Facebook, Reddit, Instagram)
  - Building sub-communities encourages trust in specific areas of interest
  - Badge/Award system to encourage positive actions.
  - Asking for feedback from user.
- Professional Services (LinkedIn, Stack Overflow)
  - Emphasizes a professional marketing environment.
  - Skill quizzes to verify reputability.

Evaluating user credibility - CredRank

- Originally used for twitter
- Graph-based.
- Users and contributions -> vertices
- Interactions -> weighted edges
- Sum of user's edge weight -> credibility score
- Interactions between users



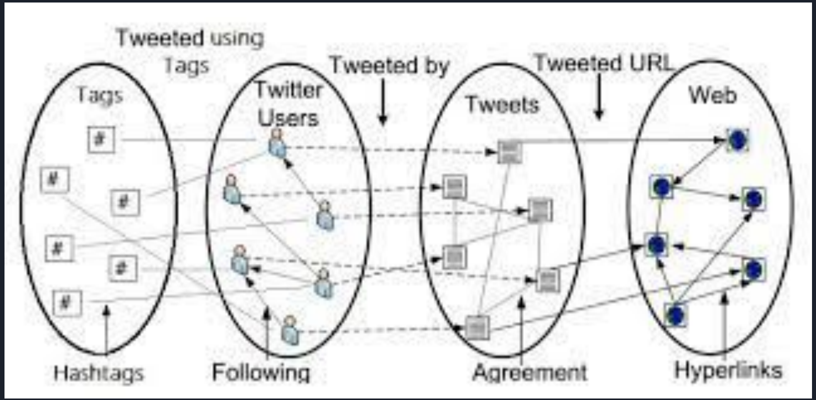
Evaluating user credibility - CredRank

Pros:

- Considers many variables
- User determined outcome
- Weights for interactions are adjustable

Cons:

- Requires existing user data
- Users/bots abusing system



Identifying Good or Bad Comments

Identify bad comments:

- Spam: irrelevant, repeated comments in large number
- Vulgarity, toxicity: explicit, offensive comments

Notify me when new comments are added

!@#$%^%!#^

Rank the remaining passable comments:

- Length: short comments are often unconstructive to the conversation
- Grammatical correctness
- Users' interaction

They're ranking have declined recently.

If I were you, I'd pick Georgia
n in engineering pr
ity, and being close to many tech hubs,
especially if you want to major in STEM.

Go to UGA, they have better football team.

Algorithm to Identify High Value Comments

The outer regression model will be used to account for all the values of features such as HVC score , User Interactions, word length, and toxicity.

An inner model will be used to generate values for HVC score and toxicity from the actual text in tokenized form. Labeled data will be used for the initial model, but for HVC, user votes will be supplemented into the model as more comments and votes are added.

HVC Score (Inner Model)

The HVC Score uses Logistic Regression NLP and current comment votes to predict future comment scores of new comments

Steps:

1. Tokenize and Preprocess data (stemming, remove stopwords)
2. TFIDF vectorization (25% testing data)
3. Account for Zero TFIDF Scores
4. Use Logistic Regression trainer
5. Generates score from 0 to 1 probability for comment being highly voted
6. Value is used in outer model

0	Username	clean	Votes	percentiles
1	ALL_GRAVY_BABY	[veri, challeng, difficult, 4, year, life, set...	159	4.0
2	pleasedonttag	[similar, post, alreadi, repli, go, love, gt, ...	61	4.0
4	Overall-Pop6295	[transfer, gt, like, 2, year, ago, person, lov...	51	4.0
0	vroomvroomski	[realli, good, time, never, depress, class, pr...	45	4.0
13	fndabs	[fr, wa, wonder, thing, offens, seem, like, co...	16	4.0
...	...	...	...	...
42	Badcomment2	[go, uga]	-10	1.0
55	Badcomment4	[nah, bro]	-15	1.0
44	Badcomment4	[nah, bro]	-15	1.0
54	Badcomment3	[want, buy, pill]	-22	1.0
43	Badcomment3	[want, buy, pill]	-22	1.0

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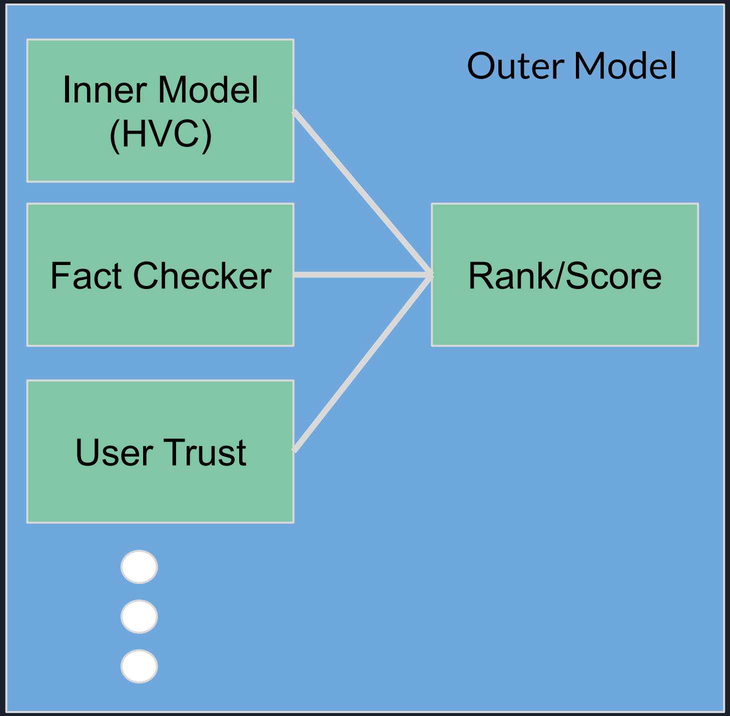
Ranking Comments (Outer Model)

Combines the Inner Model and other Parameters to rank each comment in a post using Multivariate Regression

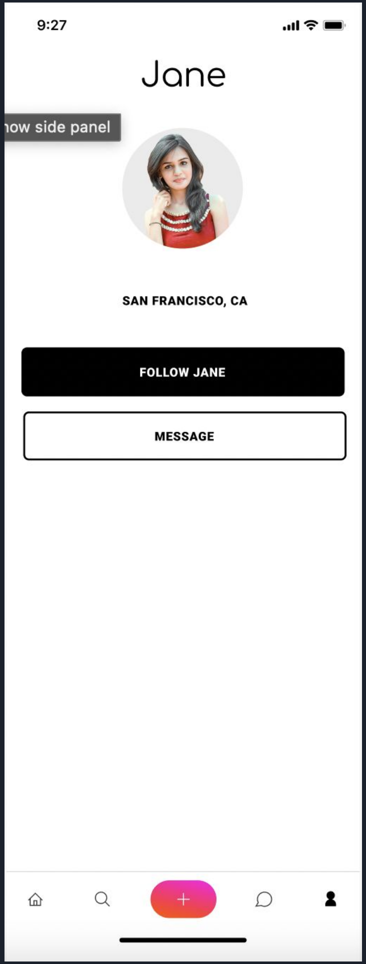
- Parameters Include
  - Fact Checker
  - Source Credibility Checker
  - User Trustworthiness Rating
  - User Knowledge in Field
    - using a set of hashtag for each user's knowledge range
  - Credibility Ratings
    - Degree
    - Certifications
    - LinkedIn Assessments
  - Similarity with Other Comments
    - Repetition vs High Value
    - Main Comment vs Repeater
  - Support/Opposition from Other Users
    - Replies (positive and negative)

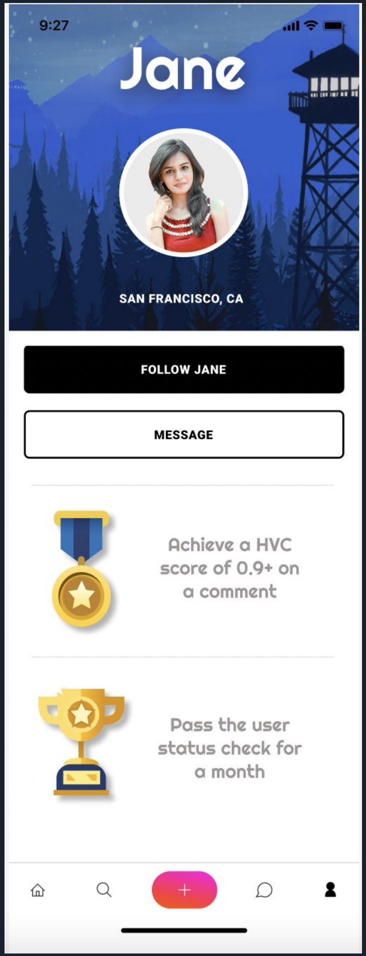


* Dismissing Comments : if a comment has a very low score, we would remove it from post for optimization reasonings

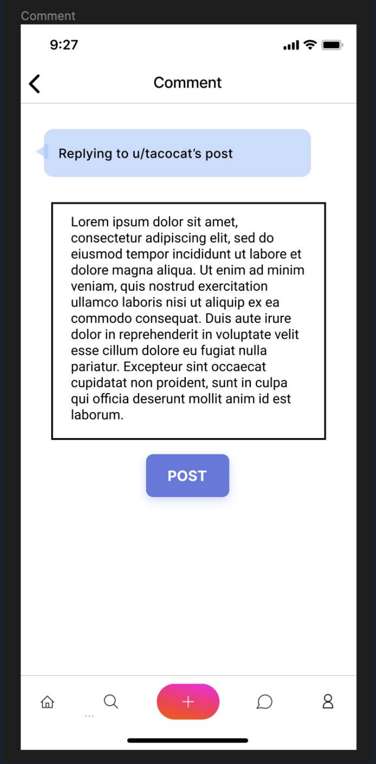
Rewarding Users

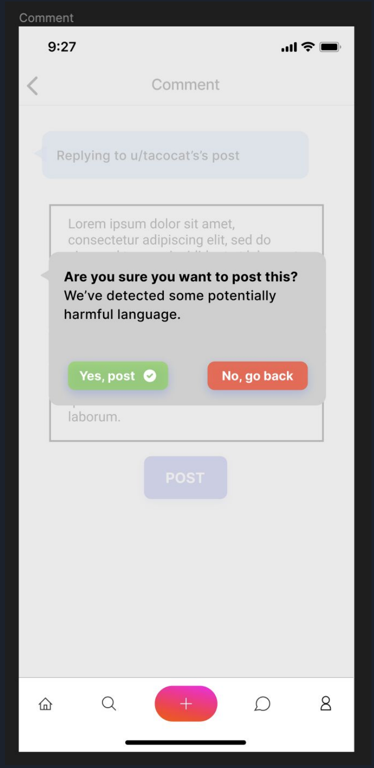
- Gamification through a hierarchical badge system (a reward system) has been shown to elicit more voluntary contributions to discussion forums.
- If users consistently contribute high value comments, they can receive badges or other benefits not available to users without these qualifications
  - For example, gif profile pictures or banners, can change usernames or add color, etc
- Rewards only produced negative effects when the rewards were expected, tangible, and provided for mere completion of a task.
  - To make badges unexpected, the criteria for earning each badge should not be displayed to the user until it has been awarded. → Users will build intrinsic motivation.
- Let’s take a look at how this could look in our UI!

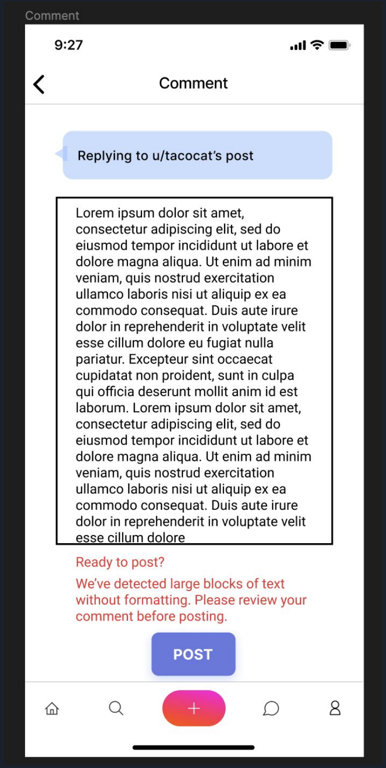

9:27
Jane
how side panel
SAN FRANCISCO, CA
FOLLOW JANE
MESSAGE


9:27
Jane
SAN FRANCISCO, CA
FOLLOW JANE
MESSAGE
Achieve a HVC score of 0.9+ on a comment
Pass the user status check for a month

Encouraging HVC







Types of “Bad” Comments

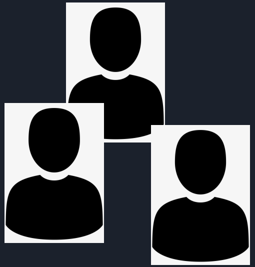
- Short and uninformative
  - Ex. “No.”, “You’re wrong.”, “You don’t know what you are talking about.”
  - Provide little information and don’t generate discussion
- Repetitive comments
  - Repeating previous comments on the thread or commenting the same thing over the course of discussion without acknowledging other items
- Bot-Generated comments
  - Bots that post the same comment repeatedly under a large number of threads
  - Significant number of bots on Instagram/Twitter
- Toxic comments
  - Consistent of foul language and personal attacks

Discouraging Bad Comments

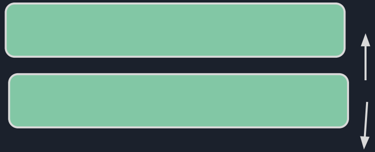
- Research shows that toxicity is due to the “nature” of a user. As a result we could try implementing a status check for every time the user logs onto the website and restrict forum access if in a “toxic” state
- Adding a cooldown timer for first-time identified “trolls” for their next comment/contribution before banning
  - Reddit implemented cool down time for in between comments
- Adding an authentication system
  - So that users can’t easily create new accounts to continue trolling

Future Plans

- Using the credibility rank algorithm and modifying it to rank a sample network of user profiles and interactions



- Identify specific criteria for and build an algorithm for generating scores to rank comments



- Implement platform features for encouraging high-value comments and discouraging low-value comments (next slide)
- The user interface models for the rewards and badge system and other main screens for the platform

Future Plans: Encouraging good comments

Possible implementation ideas/suggestions:

- Using text analysis, if an unproductive/negative comment has been detected, do not allow the user to post it yet; request for the user to review his/her comment and make any necessary modifications before posting.
- Reward the user with benefits after contributing with high-value comments
- Using an algorithm for scoring/ranking comments, reward that user if his/her comment achieves a certain threshold.
- To prevent possible innate motivation, do not reveal the reward criteria to the user until after the user has been rewarded.

Questions!